

DyMD: Preserving Interaction Dynamics through Distribution Matching Distillation in Few-Step Video World Models

Haojun Xu^{1,2} Jie Huang^{2,†} Xin Lu² Mingchen Zhong² Zihao Fan² Linjiang Huang^{1,†} Si Liu¹

¹Beihang University ²JD Future Academy

*Equal contribution. [†]Corresponding author. [‡]Project leader.

Abstract

Large video diffusion models provide rich priors for embodied prediction and learning. However, generating visual futures with these models requires costly iterative sampling. Distribution Matching Distillation (DMD) enables few-step video generation, but suppress robot-object motion while preserving visual quality. Examining DMD’s teacher and fake-score signals, we find that weak re-noising keeps the teacher posterior concentrated near motion-deficient rollouts, limiting motion-restoring guidance. Meanwhile, stronger-motion rollouts tend to incur larger fake-score fitting errors, which hinder the generator’s learning of interaction dynamics. We propose DyMD, a DMD framework that adapts both teacher supervision and critic fitting to the evolving student. Temporal affinity-conditioned re-noise sampling adapts the timestep distribution to each rollout’s current interaction fidelity by mixing the base schedule with a teacher prior motivated by local posterior variation, thereby balancing motion recovery and appearance refinement. To better track stronger-motion rollouts which are empirically associated with higher flow-matching loss, dynamics-guided fake-score tracking uses a noise-conditioned predictor to estimate noise-relative fitting difficulty, then upweights predicted-hard rollouts in the critic loss. Using DyMD, we distill a 14B teacher into a four-step 1.3B student with no auxiliary modules at inference. On embodied-video benchmarks, the student improves R-Bench task adherence by 9.6 percentage points and PAI-Bench-G Domain score by 5.1 points over Base DMD while maintaining comparable visual quality. As a backbone for downstream action planning, our student achieves 34% mean success across two WorldArena tasks, compared with 16% for Base DMD.

1. Introduction

Video world models support embodied prediction and learning by generating visual futures and providing representations for downstream policies [8, 9, 15]. For robotic manip-

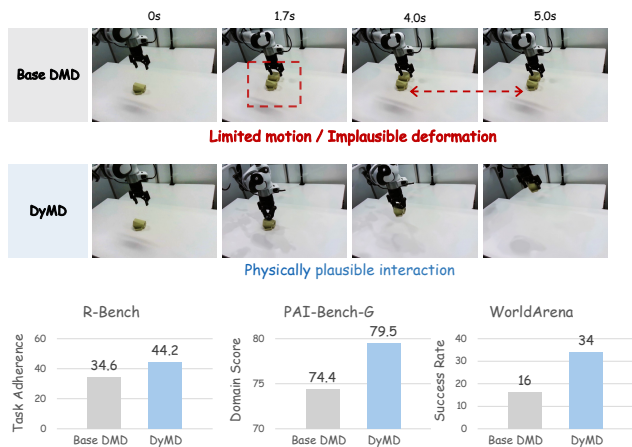


Figure 1. **Preserving interaction dynamics in few-step video world models.** Top: Base DMD shows implausible deformation and limited motion; DyMD preserves a physically plausible robot-object interaction. Bottom: gains in R-Bench task adherence, PAI-Bench-G domain score, and WorldArena planning success. Both are 1.3B four-step students.

ulation, useful predictions must capture coordinated robot-object motion as well as scene appearance. However, generating such visual futures with large video diffusion models requires costly iterative sampling [25, 30], motivating distillation into few-step generators. Distribution Matching Distillation (DMD) [32, 33] trains a few-step student using the score difference between a frozen teacher and an on-line fake-score model (the critic) that estimates the score of the evolving student distribution. But the challenge is to retain task-relevant interaction dynamics under this reduced sampling budget.

Our starting observation is that DMD preserve visual quality while suppressing interaction dynamics. When distilling PF-Wan, a teacher fine-tuned for robotic manipulation [36], vanilla DMD produces sharp videos with weak robot-object motion (Fig. 1). To understand this failure, we examine the two components of DMD’s score-difference update: the teacher guidance and the critic’s estimate of the evolving

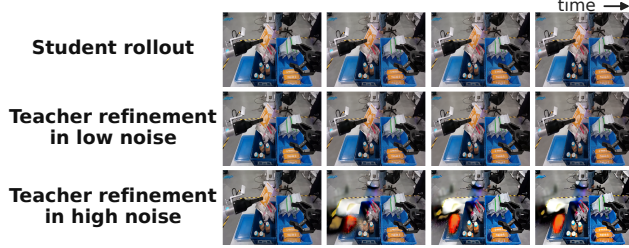


Figure 2. **Motion recovery depends on the re-noising level.** We re-noise the same nearly static four-step DMD rollout at different noise levels and refine it with the frozen PF-Wan teacher. Low re-noising levels leave the video nearly static, while higher levels recover motion but introduce visible artifacts. Four frames are shown in temporal order.

student score.

First, the teacher’s ability to correct this motion deficit depends on the re-noising level. Weak re-noising keeps the teacher’s clean-video posterior concentrated near a motion-deficient student rollout, providing little corrective guidance. Stronger re-noising of the same rollout restores robot–object motion but may introduce artifacts (Fig. 2). Typical schedules use the same timestep distribution for all rollouts, regardless of their current interaction fidelity. This motivates adapting re-noise sampling to balance motion recovery and appearance refinement for each rollout.

Second, preserving interaction dynamics requires the critic to accurately track the evolving student distribution, since errors in its student-score estimate perturb the generator’s score-difference update [14, 32]. At fixed noise levels, our frozen-snapshot probe finds that rollouts with stronger motion in VAE latent space tend to incur higher flow-matching (FM) loss (Fig. 3), suggesting that these rollouts are harder for the critic to fit. Thus, inaccurate critic scores on these rollouts hinder the generator from learning interaction dynamics effectively. To better track these stronger-motion rollouts, we therefore use fitting difficulty to allocate the critic’s limited training effort.

These two challenges motivate DyMD, which couples rollout-adaptive teacher supervision with difficulty-aware critic fitting. To balance motion recovery and appearance refinement, temporal affinity-conditioned re-noise sampling adapts each rollout’s mixture of a teacher prior and the base schedule. The prior emphasizes pronounced changes in the teacher’s denoising direction, motivated by the connection between teacher-velocity turning and local posterior variation. Temporal affinity to a paired training video, measured in frozen V-JEPA features [22], assesses the rollout’s current interaction fidelity and controls this mixture: low-affinity rollouts receive more teacher-prior sampling, while well-matched rollouts retain more base-schedule coverage. To improve the critic’s score estimates for rollouts with stronger

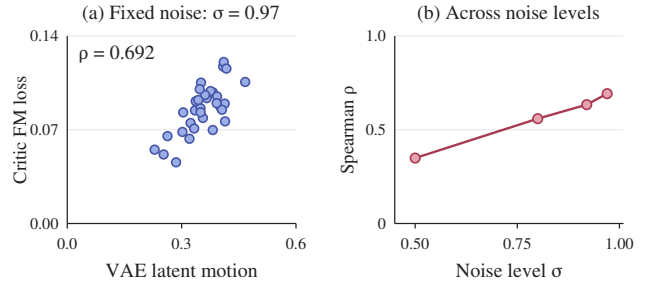


Figure 3. **Stronger latent motion is associated with higher critic FM loss.** Left: 32 frozen rollouts at $\sigma = 0.97$. Right: Spearman correlation between VAE latent motion magnitude and critic FM loss across the same rollouts, computed separately at four noise levels. Protocol in Appendix D.2.

motion, dynamics-guided fake-score tracking uses a noise-conditioned predictor to estimate noise-relative fitting difficulty from temporal dynamics in VAE latents. Rollouts predicted to be harder than typical samples at the same noise level receive larger weights in the critic loss, without increasing the number of critic updates.

We distill the PF-Wan teacher into a four-step 1.3B student. Both components operate only during training, with no auxiliary modules required at inference. Experiments on R-Bench [8], PAI-Bench-G [38], and EZS-Bench [7] show improved interaction fidelity over Base DMD: R-Bench task adherence rises by 9.6 percentage points and PAI-Bench-G Domain score by 5.1 points, with comparable visual quality. As a backbone for downstream action planning, our student achieves 34% mean success across two WorldArena tasks [28], compared with 16% for Base DMD.

Our contributions are threefold:

- We analyze motion degradation in DMD through its teacher and fake-score signals, identifying teacher-side mode locking and motion-related critic fitting difficulty as obstacles to preserving interaction dynamics.
- We propose DyMD, combining temporal affinity-conditioned re-noise sampling with dynamics-guided fake-score tracking to adapt teacher supervision and critic fitting to the current student (Sections 3.2–3.4).
- We demonstrate improved interaction fidelity over Base DMD on three embodied-video benchmarks and higher mean success on two downstream action-planning tasks with a four-step 1.3B student (Section 4).

2. Related Work

Video World Models. Large-scale video diffusion and flow models learn expressive priors for visual generation [3, 18, 25, 30]. Robot video corpora and video-based world models extend these priors to embodied prediction and learning [4, 8, 9, 15], where object persistence, con-

tact consistency, and coordinated robot–object motion are essential. VideoJAM jointly models appearance and motion to improve temporal coherence [5]. ABot-PhysWorld uses physics-aware preference optimization [7], while PhysisForcing introduces region-focused physics supervision for robotic interactions [36]. These advances improve the dynamics available in a video teacher. Our focus is transferring those dynamics to a few-step student through adaptive timestep sampling and critic fitting.

Diffusion Model Distillation. Few-step distillation transfers pretrained diffusion and flow models through trajectory consistency [21, 29] or distribution-level supervision, including adversarial learning [27] and score-based distribution matching [32, 33]. Several extensions improve the supervision used by DMD. Reward-guided methods combine distribution matching with preference optimization [16, 20], while Adaptive Matching Distillation reshapes score guidance to favor higher-quality samples [2]. For temporal fidelity, Adaptive Video Distillation combines adaptive paired regression with temporal regularization [35]. In generative video compression, Adaptive Score Distillation attenuates updates that disagree with a ground-truth reconstruction direction [31]. CoDMD adds a relational objective over samples and frames, reusing teacher and fake-score predictions to preserve their dependency structure [37].

Timestep Sampling and Fake-Score Tracking. Noise allocation and fake-score accuracy offer another axis of improvement. Improved DDPM uses loss-aware timestep importance sampling to reduce gradient noise [23], while Min-SNR balances timestep losses through SNR-based weighting [12]. Phased DMD progressively matches intermediate distributions and derives score-matching objectives within SNR subintervals for specialized experts [11]. DMD2 stabilizes fake-score estimation through two-time-scale updates [32]; RTDMD’s ambient-consistent stage adds a cross-timestep consistency regularizer to help fake-score model track the generator distribution under limited updates [14]. Learning Loss predicts sample losses for active data acquisition [34], providing a related perspective on difficulty-based allocation. Our approach uses temporal affinity to adapt each student rollout’s re-noise schedule and a noise-conditioned predictor estimates noise-relative fitting difficulty from latent temporal dynamics to reweight critic samples.

3. Method

We first review DMD’s generator and fake-score updates and the challenges they pose for preserving interaction dynamics (Section 3.1). Building on this analysis, DyMD (Fig. 4) constructs a teacher-velocity-turning prior and uses temporal affinity to adapt its mixture with the base schedule to each

rollout’s current interaction fidelity (Sections 3.2 and 3.3). It further predicts noise-relative fitting difficulty from latent temporal dynamics to reweight critic samples within a fixed update budget (Section 3.4).

3.1. Preliminaries: DMD and Its Challenges

DMD [32, 33] distills a multi-step diffusion teacher into a few-step generator G_θ , which produces a latent rollout $x_0^S = G_\theta(\xi, c)$ from noise ξ and text conditions c . It minimizes a time-averaged reverse KL between the re-noised student and teacher distributions, yielding the score-difference update

$$\nabla_\theta \mathcal{L}_G \approx \mathbb{E} \left[\left(\frac{\partial x_0^S}{\partial \theta} \right)^\top (s_\phi - s_T) \right], \quad (1)$$

where s_T is the frozen teacher score and s_ϕ is the student-score estimate provided by an online fake-score model. Both are evaluated at the re-noised rollout $x_t = (1 - \sigma_t)x_0^S + \sigma_t \varepsilon$, with noise level $\sigma_t = \sigma(t)$ and independent standard Gaussian noises ξ, ε .

Since the student score is not available in closed form, the critic learns a velocity field v_ϕ that provides this estimate through

$$s_\phi(x_t, t, c) = - \frac{x_t + (1 - \sigma_t)v_\phi(x_t, t, c)}{\sigma_t}. \quad (2)$$

With the generator fixed, v_ϕ is trained on detached student rollouts using flow matching (FM) [19]:

$$\mathcal{L}_{\text{FM}} \propto \mathbb{E} \left[\|v_\phi(x_t, t, c) - (\varepsilon - x_0^S)\|_2^2 \right]. \quad (3)$$

Through Eq. (2), FM regression provides the student-score estimate used in the generator update. DMD alternates between generator and fake-score updates, using independently sampled student rollouts for each update.

These alternating updates face two obstacles to preserving interaction dynamics.

Teacher-side mode locking: for a re-noised student rollout x_t , the teacher’s clean-video posterior satisfies:

$$p_T(x_0 | x_t, c) \propto p_T(x_0 | c) \mathcal{N}(x_t; (1 - \sigma_t)x_0, \sigma_t^2 \mathbf{I}). \quad (4)$$

At high signal-to-noise ratio $\text{SNR}_t = (1 - \sigma_t)^2 / \sigma_t^2$, the Gaussian likelihood favors clean videos near x_0^S , so the posterior may remain localized around a motion-deficient student rollout. Since the teacher score is determined by this posterior’s mean [10], even an accurate score may provide little guidance for recovering motion. Appendices C.1–C.2 relate the score difference to posterior means and analyze this mode locking in a two-mode model. Stronger re-noising relaxes this local constraint but weakens appearance evidence, motivating re-noise sampling adapted to each rollout’s current interaction fidelity.

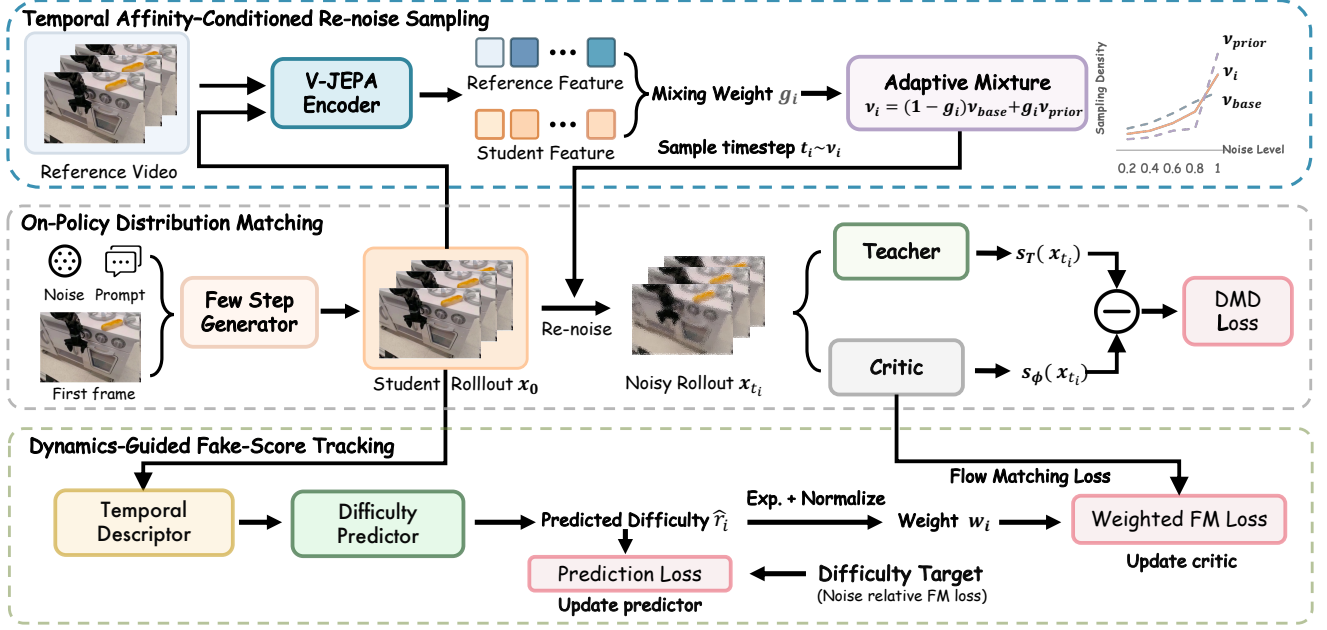


Figure 4. **Overview of DyMD.** Top (Secs. 3.2–3.3): temporal affinity controls each rollout’s mixture of the teacher prior and base schedule. Middle (Sec. 3.1): the teacher–critic score difference updates the student. Bottom (Sec. 3.4): a dynamics- and noise-conditioned predictor assigns mean-one weights to critic FM losses without changing their targets. Generator and critic share the timestep sampling rule; all auxiliary modules are training-only.

Critic lag: the score estimate s_ϕ must track a changing student distribution. With finitely many updates, v_ϕ may remain inaccurate; Eq. (2) passes this error into the generator’s score difference [11, 14, 32]. This motivates allocating limited critic fitting effort to rollouts that are harder to track.

3.2. Teacher-Velocity Turning Prior Schedule

Stronger re-noising can recover motion but may also introduce artifacts (Fig. 2). To guide timestep selection, we seek noise intervals where the teacher’s denoising direction undergoes pronounced reorganization. Let $\hat{v}_T(x_\sigma, \sigma, c)$ be its sampled velocity. The operator C centers the velocity field by subtracting the future-block temporal mean. Along trajectory m , the centered velocity and its unit direction are

$$v_m^c(\sigma) = C\hat{v}_T(x_{m,\sigma}, \sigma, c_m), \quad u_m(\sigma) = \frac{v_m^c(\sigma)}{\|v_m^c(\sigma)\|_2}. \quad (5)$$

We measure the change in teacher velocity direction per unit noise level between adjacent sampling steps:

$$\hat{\kappa}_{m,k} = \frac{\arccos(u_m(\sigma_k)^\top u_m(\sigma_{k+1}))}{|\sigma_{k+1} - \sigma_k|}. \quad (6)$$

Its continuous limit is $\kappa_m(\sigma) = \|du_m/d\sigma\|_2$. Small turning indicates a locally stable direction; large turning marks pronounced directional reorganization.

Posterior interpretation. To connect turning to clean-video predictions, use the conditional field v_T in Eq. (5). Along trajectory m , the clean posterior and its mean satisfy

$$\begin{aligned} \pi_{m,\sigma}(x_0) &= p_T(x_0 | x_{m,\sigma}, c_m), \\ \mu_{m,\sigma} &= x_{m,\sigma} - \sigma v_T(x_{m,\sigma}, \sigma, c_m). \end{aligned} \quad (7)$$

Along a flow trajectory, $dx/d\sigma = v_T$ implies $d\mu/d\sigma = -\sigma dv_T/d\sigma$; thus velocity turning requires a change in the posterior mean. For neighboring noise levels, the local KL expansion [24] measures variation of the full posterior:

$$D_{\text{KL}}(\pi_{m,\sigma} \| \pi_{m,\sigma+\delta}) = \frac{1}{2}\mathcal{I}_m(\sigma)\delta^2 + o(\delta^2), \quad (8)$$

where $\mathcal{I}_m(\sigma)$ is the posterior-path Fisher information, quantifying local posterior sensitivity to σ along the trajectory. The following result links this posterior variation to turning.

Proposition 1 (Turning witnesses posterior variation). *For regular posterior paths of the exact conditional flow, suppose $\sigma\|v_m^c(\sigma)\|_2$ is uniformly bounded away from zero. Suppose the centered posterior covariance, projected orthogonally to u_m , has uniformly bounded trace. Then a constant $C_I > 0$ exists such that*

$$\mathbb{E}_m[\mathcal{I}_m(\sigma)] \geq C_I \bar{\kappa}(\sigma)^2, \quad \bar{\kappa}(\sigma) = \mathbb{E}_m[\kappa_m(\sigma)]. \quad (9)$$

The bound motivates the squared mean turning rate as a geometric indicator of posterior variation (proof in Appendix C), supporting a sampling prior that emphasizes noise

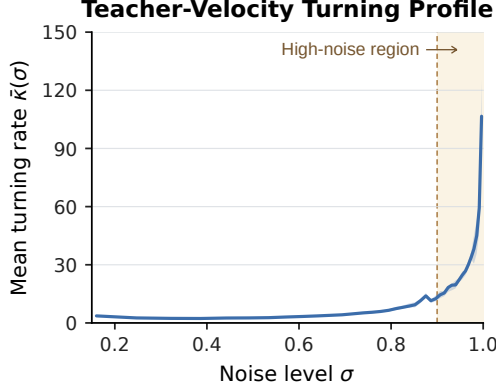


Figure 5. **Teacher-velocity turning profile.** Mean turning rate of future-block-centered velocity across PF-Wan teacher trajectories. The light-blue band shows the interquartile range across trajectories; pale-yellow shading highlights the high-noise region $\sigma \in [0.9, 1]$.

intervals with pronounced denoising-direction reorganization. In practice, we profile 32 trajectories from the PF-Wan teacher [36] using its official 40-step solver with a CFG scale of 5. The measured turning rates are first averaged across trajectories to estimate $\bar{\kappa}(\sigma)$, then smoothed over σ to obtain $\tilde{\kappa}(\sigma)$ (Fig. 5). The teacher-velocity-turning prior schedule is defined on $\mathcal{T}$, the timestep range used for re-noising:

$$h_{\kappa}(t) = (\tilde{\kappa}(\sigma_t) + \epsilon_{\kappa})^2, \quad \nu_{\kappa}(t) = \frac{h_{\kappa}(t)}{\int_{\mathcal{T}} h_{\kappa}(u) du}. \quad (10)$$

The offset $\epsilon_{\kappa} > 0$ preserves support.

3.3. Temporal Affinity–Conditioned Re-noise Sampling

The teacher prior ν_{κ} emphasizes noise intervals with pronounced denoising-direction reorganization, but is fixed across rollouts. We adapt its mixture with the base schedule ν_{base} to each rollout’s current interaction fidelity. To assess this fidelity, we compare multi-scale temporal changes between the decoded student video V_i^S and its paired target V_i^* in a frozen video encoder’s feature space. Their pooled features and temporal differences are

$$Z_i^a = \text{Pool}(\Phi(V_i^a)), \quad a \in \{S, \star\}, \quad (11)$$

$$(\Delta_{\ell} Z_i^a)_{j,u} = (Z_i^a)_{j+\ell,u} - (Z_i^a)_{j,u},$$

where Φ is the frozen encoder, instantiated as V-JEPA 2.1 [22], and Pool normalizes and pools tokens within a 3×3 spatial grid. The indices j, u denote time and spatial cells, and $\ell \in \{1, 2, 4\}$ is the temporal lag. For brevity, write $\delta z_i^a = (\Delta_{\ell} Z_i^a)_{j,u}$ and let $\langle \cdot \rangle$ denote lag-weighted averaging over time–cell pairs. We use their *temporal affinity* A_i as a feature-space proxy for interaction fidelity:

$$A_i = \frac{2 \langle [(\delta z_i^S)^{\top} \delta z_i^{\star}]_+ \rangle}{\langle \|\delta z_i^S\|_2^2 + \|\delta z_i^{\star}\|_2^2 \rangle}, \quad (12)$$

where $[\cdot]_+ = \max(\cdot, 0)$ and $A_i \in [0, 1]$. Identical temporal changes give one, while mismatches in direction or magnitude reduce affinity.

The affinity sets the teacher-prior probability g_i and timestep sampling density:

$$g_i = \frac{\alpha}{\alpha + (1 - \alpha)A_i}, \quad 0 < \alpha \leq 1, \quad (13)$$

$$\nu_i(t) = (1 - g_i)\nu_{\text{base}}(t) + g_i\nu_{\kappa}(t).$$

Here α sets the minimum teacher-prior probability. Low-affinity rollouts receive more teacher-prior sampling, while well-matched rollouts retain more base-schedule coverage for appearance refinement. Both generator and critic sample $t_i \sim \nu_i$, with ν_i computed from their respective rollouts. This aligns critic training with the noise-level allocation used for generator updates.

3.4. Dynamics-Guided Fake-Score Tracking

Even at a fixed noise level, critic fitting difficulty varies across rollouts: stronger latent motion is associated with higher FM loss (Fig. 3). We therefore use latent temporal dynamics to predict noise-relative fitting difficulty and allocate fitting effort within a fixed critic-update budget.

To compare rollouts evaluated at different noise levels, we center log-FM loss by a noise-bin baseline $b_{k(t_i)}$, an exponential moving average of log-FM loss in the bin containing t_i . We define the *noise-relative fitting difficulty* as

$$r_i = \text{sg}[\log \ell_i^{\text{FM}} - b_{k(t_i)}]. \quad (14)$$

A positive r_i indicates higher log-FM loss than the running baseline at a comparable noise level.

We predict this difficulty from VAE latent dynamics, measured in the critic’s FM regression space. For a detached student latent $X_i = \text{sg}[x_{0,i}^S]$, a fixed map $\mathcal{D}_{\text{temp}}$ aggregates RMS magnitudes of multi-lag latent differences and displacements from the first block using temporal statistics such as the mean and standard deviation. A lightweight MLP f maps the descriptor d_i and a noise-level embedding $e(t_i)$ to the predicted fitting difficulty $\hat{r}_i$:

$$d_i = \mathcal{D}_{\text{temp}}(X_i), \quad (15)$$

$$\hat{r}_i = f(d_i, e(t_i)).$$

We train f online with a Huber loss $\mathcal{L}_{\text{pred}}$ against r_i . Descriptor details are given in Appendix E.1.

To emphasize predicted-hard rollouts within the existing critic updates, we exponentiate $\hat{r}_i$ and normalize the weights to unit mean over the global minibatch of size B :

$$w_i = \frac{\exp(\beta \hat{r}_i)}{B^{-1} \sum_{j=1}^B \exp(\beta \hat{r}_j)}. \quad (16)$$

Table 1. **Main comparison on embodied-video benchmarks.** The teacher and undistilled student are reference models and are separated from the two matched four-step DMD variants. Bold with blue shading marks the better result within the matched pair. NFE counts denoiser evaluations, including both CFG branches. Benchmark scores are percentages and higher is better. R-Bench embodiment columns are abbreviated as Dual (dual-arm), Hum. (humanoid), Single (single-arm), and Quad. (quadruped).

Method	Params	NFE	R-Bench: TAC					PAI-Bench-G					EZS-Bench			
			Overall	Dual	Hum.	Single	Quad.	Domain	Space	Physics	Time	Quality	Domain	Space	Physics	Time
PF-Wan	14B	80	75.3	72.5	79.8	71.8	77.2	89.0	91.8	92.2	88.4	76.9	83.2	80.9	89.2	78.0
Wan2.1-Fun	1.3B	100	47.3	42.5	53.2	30.8	63.2	79.0	79.7	82.5	76.4	75.3	75.5	71.7	83.9	68.8
Base DMD	1.3B	4	34.6	25.2	33.4	25.8	54.0	74.4	77.0	82.8	64.7	78.1	73.3	66.9	86.0	64.8
DyMD	1.3B	4	44.2	38.8	48.0	31.2	58.8	79.5	82.4	81.2	73.5	78.1	76.2	71.8	84.4	70.8



Figure 6. **Visual comparison of interaction dynamics between Base DMD and DyMD at four NFE.** Uniformly spaced frames from two PAI-Bench cases. Top: lifting a blue shipping container. Bottom: handing a menu to a person. Base DMD fails to lift the target container and produces limited handover motion, whereas DyMD realizes the instructed interaction.

Here $\beta > 0$ controls the strength on predicted-hard rollouts. The training objective is

$$\mathcal{L}_C = \frac{1}{B} \sum_{i=1}^B \text{sg}[w_i] \ell_i^{\text{FM}}(\phi) + \lambda_{\text{pred}} \mathcal{L}_{\text{pred}}. \quad (17)$$

Rollouts with higher predicted fitting difficulty receive larger coefficients in the critic gradient. Detached weights ensure that f learns only from $\mathcal{L}_{\text{pred}}$ and all auxiliary modules are training-only.

4. Experiments

4.1. Experimental Setup

Models and training. We distill the PF-Wan 14B image-to-video teacher [36] into a bidirectional 1.3B student. PF-Wan is fine-tuned on robotic manipulation videos with interaction-focused physics supervision, making it a suitable teacher for transferring interaction dynamics to a few-step student. The student and fake-score model are initialized

from Wan2.1-Fun-V1.1-1.3B-InP [1]. The teacher uses its official 40-step, shift-5 sampler with CFG 5; the student generates 81-frame videos in four denoiser evaluations. A frozen V-JEPA 2.1-L encoder provides temporal-affinity features. We train on 32,897 clips curated from RoVid-X [8] through semantic deduplication, image-text alignment, and motion/visibility filtering. All DMD variants use the same 2,500-step training budget and effective batch size of 32. Tracking reweights samples without changing the number of critic updates. We set the teacher-prior probability floor to $\alpha = 0.3$ and critic-weighting strength to $\beta = 0.5$. Data curation and implementation details are provided in Appendices A and B, with hyperparameters listed in Table S3.

Evaluation. For in-domain evaluation, we report Task-Adherence Consistency (TAC) on 400 held-out image-prompt pairs from R-Bench [8], spanning four robot embodiments. We assess cross-dataset generalization on the robot subset of PAI-Bench-G [38] (174 samples) and EZS-Bench [7] (196 samples), reporting the Domain score along-

Table 2. **Sampling and fake-score tracking ablation.** Sampling denotes temporal affinity-conditioned re-noise sampling; Tracking denotes dynamics-guided critic weighting. All variants retain the critic and use the same update and inference budgets. Bold with blue shading marks the best result in each column.

Method	Components		R-Bench	PAI-Bench-G		EZS-Bench
	Sampling	Tracking	TAC	Domain	Quality	Domain
Base DMD	×	×	34.6	74.4	78.1	73.3
w/o Tracking	✓	×	43.5	78.5	78.1	75.1
w/o Sampling	×	✓	38.3	75.4	78.0	74.0
DyMD	✓	✓	44.2	79.5	78.1	76.2

side Space, Physics, and Time scores. We also report the official PAI-Bench-G Quality score to assess visual quality. All models are evaluated with the same initial images, prompts, and random seeds. For EZS-Bench, we follow PhysisForcing [36] in compressing the original descriptions into action-centric prompts for video generation. Detailed evaluation protocols are provided in Appendix F.

4.2. Main Results

As shown in Table 1, our student uses 4 NFE, versus 80 for the teacher and 100 for the undistilled student. Base DMD, which uses the base re-noise schedule and uniform critic weights, substantially degrades interaction fidelity: R-Bench TAC falls to 34.6 and PAI Domain to 74.4, despite an increase in PAI Quality from 75.3 to 78.1. These results indicate that Base DMD preserves visual quality but does not adequately retain task-relevant interaction dynamics.

At the same four-NFE budget, DyMD better preserves interaction dynamics, raising R-Bench TAC to 44.2 (+9.6 points), and PAI/EZS Domain to 79.5/76.2 (+5.1/ + 2.9 points). Base DMD scores 1.6 points higher on Physics in PAI-Bench-G because its near-static outputs preserve plausible physical states without executing the instructed interaction (Fig. 6).

4.3. Representation Analysis for the Generator and Critic

We compare VAE and V-JEPA representations for two training signals: target-relative temporal affinity for generator sampling and noise-relative fitting difficulty for critic tracking. Each probe uses a frozen 32-rollout bank.

Generator representations. We compute temporal affinity A_i (Section 3.3) in both spaces. As an independent motion reference, we compute a *trajectory agreement* score T_i using point trajectories tracked by CoTracker3 [17]. This score measures how closely the generated video matches the target video in the directions and magnitudes of point displacements. Figure 7(a) compares the rankings induced by A_i and T_i . V-JEPA achieves a higher Spearman correlation

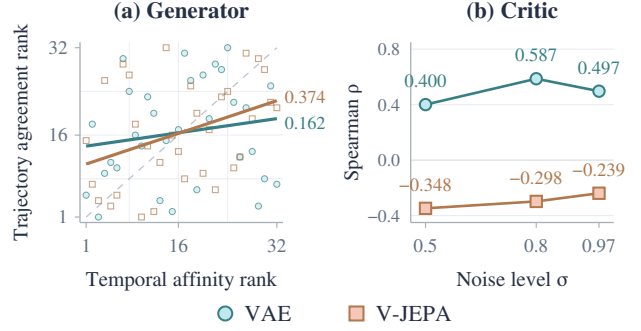


Figure 7. **Representations for generator sampling and critic tracking.** (a) Rank comparison between temporal affinity and CoTracker3-based trajectory agreement across 32 student-target video pairs. Solid lines show linear fits and the dashed diagonal denotes identical rankings. (b) Spearman ρ between MLP predictions and log-FM loss, computed separately at each noise level on a 32-rollout bank. FM losses are averaged over three independent noise draws per rollout and noise level.

with trajectory agreement than VAE (0.374 vs. 0.162), indicating that its temporal affinity better reflects target-relative motion consistency on these rollouts.

Critic representations. We fit small offline MLPs to predict log-FM loss from temporal descriptors and evaluate them on held-out rollouts. Figure 7(b) reports Spearman correlations computed separately at each noise level $\sigma \in \{0.5, 0.8, 0.97\}$. VAE-based predictions are positively correlated with the measured loss across all tested noise levels, whereas V-JEPA-based predictions are consistently negatively correlated. At a fixed noise level, subtracting the shared baseline preserves the ranking of log-FM losses, so this probe evaluates the sample ordering for noise-relative fitting difficulty. These results support VAE descriptors for critic tracking, complementing V-JEPA features for target-relative motion assessment. Appendix D.3 evaluates the trained online predictor. Its predictions remain strongly correlated with measured FM loss, supporting its ability to distinguish samples by critic fitting difficulty.

4.4. Ablation Study

Re-noise sampling and fake-score tracking. Table 2 evaluates the individual and combined effects of two components. Sampling alone raises R-Bench TAC/PAI Domain from 34.6/74.4 to 43.5/78.5, while tracking alone achieves 38.3/75.4. Adding tracking to adaptive sampling further improves R-Bench TAC/PAI Domain to 44.2/79.5 and EZS Domain from 75.1 to 76.2, with PAI Quality unchanged at 78.1. These results show that fake-score tracking complement adaptive re-noise sampling.

Table 3. **Re-noise sampling-schedule ablation.** We compare Base DMD with a high-noise control (Shift-25, raising the flow shift from 5 to 25), the teacher prior, and our temporal affinity-conditioned sampler. All variants disable critic reweighting ($\beta = 0$). Bold with blue shading marks the best result in each column.

Sampling schedule	R-Bench	PAI-Bench-G		EZS-Bench
	TAC	Domain	Quality	Domain
Base DMD	34.6	74.4	78.1	73.3
Shift-25	40.0	77.2	78.2	74.6
Teacher prior	42.5	77.9	78.0	75.0
Affinity-conditioned	43.5	78.5	78.1	75.1

Table 4. **Sensitivity to critic-weighting strength β .** We vary only β , with $\alpha = 0.3$. The $\beta = 0.5$ row repeats the full-model results from Table 1. Bold with blue shading marks column-wise best values, including ties.

β	R-Bench	PAI-Bench-G		EZS-Bench
	TAC	Domain	Quality	Domain
0.1	43.8	79.1	78.0	75.4
0.5	44.2	79.5	78.1	76.2
1.0	43.6	78.4	78.1	75.1

Re-noise sampling schedules. Table 3 compares re-noise schedules with critic reweighting disabled ($\beta = 0$). We choose Shift-25 to approximately match the teacher prior’s probability mass at $\text{SNR} < 1$ (Appendix B.3). Increasing the shift from 5 to 25 raises R-Bench TAC/PAI Domain from 34.6/74.4 to 40.0/77.2, supporting stronger re-noising for motion recovery. The teacher prior further improves them to 42.5/77.9 at similar total high-noise mass, supporting timestep sampling informed by the teacher’s flow geometry across noise levels. Affinity conditioning achieves the best R-Bench TAC and Domain scores (43.5/78.5/75.1), supporting rollout-specific adaptation beyond a fixed prior.

Critic-weighting strength. At fixed $\alpha = 0.3$ (Table 4), both $\beta = 0.1$ and 0.5 improve TAC and both Domain scores over uniform critic weighting, supporting moderate emphasis on motion-related hard rollouts. Increasing β from 0.5 to 1.0 reduces R-Bench TAC and Domain scores, suggesting that excessive emphasis on hard rollouts compromise score estimation on typical samples under a fixed update budget.

4.5. Qualitative Results

As illustrated in Fig. 6, base DMD often produces nearly static videos or limited gripper motion without completing the intended interaction. In contrast, DyMD generates coordinated robot-object motion while preserving scene appearance. In the container-lifting example, Base DMD moves the lifting mechanism without lifting the target container,

Table 5. **WorldArena action-planner success rate (%)**. Upper block: results reported in PhysisForcing [36]. Lower block: our 1.3B backbones with separately trained, task-specific VPP heads. Bold with blue shading compares the lower block only.

Model	Adjust Bottle	Click Bell	Avg.
Genie Envisioner	10.0	20.0	15.0
Tesseract	1.0	35.0	18.0
RoboMaster	8.0	20.0	14.0
Vidar	2.0	19.0	10.5
WoW	20.0	21.0	20.5
Wan2.2-Ti2V-5B	12.0	20.0	16.0
Base DMD (Wan2.1-1.3B)	1.0	31.0	16.0
DyMD (Wan2.1-1.3B)	2.0	66.0	34.0

whereas DyMD lifts the blue container from the stack. In the menu-handover example, Base DMD moves the grippers but fails to complete the handover, whereas DyMD depicts the robot successfully passing the menu to the person.

4.6. Downstream Action Planning

Following WorldArena action-planner protocol [28], we compare Base DMD and DyMD as frozen 1.3B backbones on Adjust Bottle and Click Bell in RoboTwin 2.0 [6]. For each frozen 1.3B backbone, we train task-specific VPP heads [13] to predict actions from intermediate DiT features, using 50 demonstrations per task for 200 epochs. We report closed-loop success rates over 100 episodes per task. Implementation details are in Appendix F.1.

Using DyMD as the backbone raises mean success from Base DMD’s 16.0% to 34.0% (Table 5). The gain comes mainly from Click Bell (31.0% $\rightarrow$ 66.0%), with a smaller increase on Adjust Bottle (1.0% $\rightarrow$ 2.0%).

5. Conclusion

We presented DyMD for preserving interaction dynamics in few-step video world models. Temporal affinity-conditioned re-noise sampling adapts the mixture of the teacher-velocity-turning prior and the base schedule to each rollout’s current interaction fidelity. To improve score estimation for rollouts with stronger motion, dynamics-guided fake-score tracking reweights the critic loss according to predicted noise-relative fitting difficulty. The resulting four-step 1.3B student improves R-Bench TAC and Domain scores on PAI-Bench-G and EZS-Bench over Base DMD without auxiliary inference modules. As a frozen backbone for downstream action planning, the student also improves the mean success rate from Base DMD’s 16% to 34% across two WorldArena tasks.

Limitations. The turning profile is teacher- and schedule-specific and must be re-estimated for a new teacher.

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DyMD: Preserving Interaction Dynamics through Distribution Matching Distillation in Few-Step Video World Models

Supplementary Material

Stage	Criterion	Input	Output
Deduplication	greedy cover, 0.95	2,824,039	448,528
Image-text	top-50% cosine	448,528	224,264
Motion	tracks ≥ 50 , vis. ≥ 0.6	224,264	32,897

Table S1. Training-data curation funnel.

A. Training-Set Curation

We curate a training set of 32,897 clips from the 2,824,039-clip RoVid-X corpus [8] through three stages, prioritizing semantic diversity and measurable interaction motion.

Stage 1: semantic caption deduplication. We first remove exact duplicates from the `short_caption` field, obtaining 1,374,834 distinct captions. We then encode these captions with a ViT-L/14 text encoder and ℓ_2 -normalize the embeddings. To remove semantically redundant captions, we apply a greedy covering procedure with a cosine similarity threshold of 0.95. We process captions in descending order of frequency, retaining each remaining caption as a representative and discarding all remaining captions whose similarity to it is at least 0.95. Unlike connected-component clustering, this procedure avoids merging captions through chains of intermediate matches and retains 448,528 representatives.

Stage 2: image-text filtering. For each retained clip, we decode the first frame and compute its similarity to the caption using ViT-L/14 image and text embeddings. We retain the top 50% by similarity, leaving 224,264 clips. This stage filters visual-textual mismatches that caption-only deduplication cannot detect.

Stage 3: motion and visibility filtering. We track points with CoTracker3 [17] and retain clips with at least 50 selected tracks and a mean visibility of at least 0.6. This retains 14.7% of the remaining clips, yielding the final training set of 32,897 clips.

B. Implementation Details

B.1. Image-to-video conditioning

The 1.3B student and critic use the Wan2.1-Fun I2V interface [1] and share the 16-channel Wan2.1 latent space with the PF-Wan teacher. For image conditioning, we construct a video containing the input image as its first frame and zeros

in all subsequent frames, then encode it with the Wan2.1 VAE. An accompanying mask marks only the first latent temporal block as observed.

The student performs four bidirectional denoising evaluations at model times $\{1, 0.75, 0.5, 0.25\}$ with flow shift 5. It generates 21 latent temporal blocks, decoded to 81 frames at 16 FPS. The teacher uses its official 40-step, shift-5, CFG-5 schedule. During DMD, teacher-score evaluations use CFG 5; the critic is conditional without CFG.

B.2. Video features for temporal affinity

We sample 16 RGB frames uniformly over the 81-frame video. V-JEPA produces an $8 \times 24 \times 32 \times 1024$ token grid. We normalize its tokens, pool them within a fixed 3×3 full-frame grid, and normalize the pooled features. Paired targets are cached as $8 \times 9 \times 1024$ fp16 features.

B.3. Teacher prior schedule and re-noise sampling

The base DMD schedule samples raw model time uniformly and applies the flow-shift transformation with shift 5. For $u \sim \mathcal{U}[0, 1)$, the shift and physical noise coordinate are

$$\sigma_t = t = f_s(u) = 1000 \frac{s(u)}{1 + (s-1)(u)} \quad (18)$$

We sample re-noising timesteps within $\mathcal{T} = [0.020, 0.999)$. For $s > 1$, the shift moves raw times toward higher noise.

For the Shift-25 control in Table 3, we choose $s = 25$ to approximately match the teacher prior’s mass in the high-noise region $\text{SNR}_t < 1$. On the common interval $\mathcal{T} = [0.020, 0.999)$, the normalized CDFs give probabilities of 96.14% for Shift-25 and 95.69% the teacher prior.

C. Geometry and Posterior Interpretation of Teacher-Velocity Turning

We relate DMD’s score difference to posterior means, analyze high-SNR mode locking in a two-mode model, and prove the connection between teacher-velocity turning and posterior variation.

C.1. Score difference and posterior means

Let $q_{\theta,t}$ and $p_{T,t}$ denote the noisy marginals obtained by applying the re-noising process in Section 3.1 to the clean student and teacher distributions, with exact scores $s_{q,t}$ and s_T . Let $\mu_{q,t}$ and $\mu_{T,t}$ be the corresponding clean posterior

means conditioned on the same (x_t, c) . Gaussian conditioning gives

$$s_{q,t} - s_T = \frac{1 - \sigma_t}{\sigma_t^2} (\mu_{q,t} - \mu_{T,t}). \quad (19)$$

Thus, the exact score difference depends on the two posterior means; the analysis below shows how high SNR can cause the teacher posterior to concentrate near a motion-deficient student rollout.

C.2. Mode-Locking Analysis

Proposition 2 (High-SNR mode locking). *Consider a teacher with two clean modes $p_{T,0} = \pi_s \delta_{x_s} + \pi_m \delta_{x_m}$, where δ_x is a Dirac measure, $\pi_s, \pi_m > 0$, $\pi_s + \pi_m = 1$, x_s is a static mode, and x_m is a moving mode. Re-noise a collapsed student sample as $x_t = (1 - \sigma_t)x_s + \sigma_t \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$, and let $d = x_m - x_s$ and $\text{SNR}_t = (1 - \sigma_t)^2 / \sigma_t^2$. The teacher posterior odds and their expected log value satisfy*

$$\begin{aligned} \log \frac{p_T(x_m | x_t, c)}{p_T(x_s | x_t, c)} &= \log \frac{\pi_m}{\pi_s} - \frac{\text{SNR}_t}{2} \|d\|_2^2 \\ &\quad + \frac{1 - \sigma_t}{\sigma_t} \langle \varepsilon, d \rangle, \quad (20) \\ \mathbb{E}_\varepsilon \left[\log \frac{p_T(x_m | x_t, c)}{p_T(x_s | x_t, c)} \right] &= \log \frac{\pi_m}{\pi_s} - \frac{\text{SNR}_t}{2} \|d\|_2^2. \end{aligned}$$

The expected log-odds of the moving mode decrease with SNR and squared mode separation, favoring the static mode at high SNR.

Proof. Bayes' rule reduces the posterior odds to the prior odds times the ratio of the two Gaussian likelihoods. Expanding their squared distances gives Eq. (20). $\square$

C.3. Posterior Variation and Proof of Proposition 1

Setup. For L latent temporal blocks $v = (v_1, \dots, v_L)$, define the linear operator

$$Hv = \text{vec} \left[v_j - \frac{1}{L-1} \sum_{\ell=2}^L v_\ell \right]_{j=2}^L. \quad (21)$$

The operator H excludes the observed block and subtracts the temporal mean over the future blocks.

We consider the linear noising process $x_\sigma = (1 - \sigma)x_0 + \sigma\varepsilon$, where ε is standard Gaussian noise independent of x_0 . Let v_T denote its exact conditional velocity field. Along a characteristic $x_{m,\sigma}$ satisfying $dx_{m,\sigma}/d\sigma = v_T(x_{m,\sigma}, \sigma, c_m)$, define

$$\begin{aligned} \pi_{m,\sigma}(x_0) &= p_T(x_0 | x_{m,\sigma}, c_m), \\ \mathcal{I}_m(\sigma) &= \mathbb{E}_{\pi_{m,\sigma}} \left[\left(\frac{d}{d\sigma} \log \pi_{m,\sigma}(x_0) \right)^2 \right]. \quad (22) \end{aligned}$$

Here the derivative follows the trajectory $x_{m,\sigma}$. Write $\mu_{m,\sigma} = \mathbb{E}_{\pi_{m,\sigma}}[x_0]$, $\Sigma_{m,\sigma} = \text{Cov}_{\pi_{m,\sigma}}(x_0)$, and $v_{m,\sigma} =$

$v_T(x_{m,\sigma}, \sigma, c_m)$. Define the centered velocity and its direction as

$$v_m^c(\sigma) = H v_{m,\sigma}, \quad u_m(\sigma) = \frac{v_m^c(\sigma)}{\|v_m^c(\sigma)\|_2},$$

with $P_{m,\sigma} = \mathbf{I} - u_m u_m^\top$ and $\kappa_m(\sigma) = \|\dot{u}_m(\sigma)\|_2$. Dots denote total derivatives along the characteristic. We assume a regular posterior path, $\sigma > 0$, nonzero centered velocity, and a positive projected covariance trace.

Pointwise bound. For the linear noising process, the exact conditional velocity satisfies

$$\begin{aligned} v_{m,\sigma} &= \mathbb{E}_T[\varepsilon - x_0 | x_{m,\sigma}, c_m] \\ &= \frac{x_{m,\sigma} - \mu_{m,\sigma}}{\sigma}. \quad (23) \end{aligned}$$

Differentiating $\mu_{m,\sigma} = x_{m,\sigma} - \sigma v_{m,\sigma}$ along the characteristic gives

$$\dot{\mu}_{m,\sigma} = -\sigma \dot{v}_{m,\sigma}. \quad (24)$$

Since H is fixed, differentiation of the normalized centered velocity yields

$$\dot{u}_m = \frac{P_{m,\sigma} H \dot{v}_{m,\sigma}}{\|H v_{m,\sigma}\|_2}. \quad (25)$$

Combining this identity with Eq. (24) gives

$$\kappa_m(\sigma) = \frac{\|P_{m,\sigma} H \dot{\mu}_{m,\sigma}\|_2}{\sigma \|H v_{m,\sigma}\|_2}. \quad (26)$$

To relate posterior-mean variation to $\mathcal{I}_m(\sigma)$, define the posterior-path score

$$\ell_{m,\sigma}(x_0) = \frac{d}{d\sigma} \log \pi_{m,\sigma}(x_0). \quad (27)$$

Normalization gives $\mathbb{E}_{\pi_{m,\sigma}}[\ell_{m,\sigma}] = 0$. Differentiating the posterior mean under the integral therefore yields

$$\dot{\mu}_{m,\sigma} = \mathbb{E}_{\pi_{m,\sigma}}[(x_0 - \mu_{m,\sigma}) \ell_{m,\sigma}(x_0)]. \quad (28)$$

Applying $P_{m,\sigma} H$ and using Cauchy–Schwarz gives

$$\begin{aligned} \|P_{m,\sigma} H \dot{\mu}_{m,\sigma}\|_2^2 &\leq \text{tr}(P_{m,\sigma} H \Sigma_{m,\sigma} H^\top P_{m,\sigma}) \mathcal{I}_m(\sigma). \quad (29) \end{aligned}$$

Substituting Eq. (26) and rearranging establishes the pointwise bound:

$$\mathcal{I}_m(\sigma) \geq \frac{\sigma^2 \|v_m^c(\sigma)\|_2^2 \kappa_m(\sigma)^2}{\text{tr}(P_{m,\sigma} H \Sigma_{m,\sigma} H^\top P_{m,\sigma})}. \quad (30)$$

Average bound. Suppose there exist constants $c_v > 0$ and $0 < C_\Sigma < \infty$ such that, uniformly over the trajectories and noise interval under consideration,

$$\begin{aligned} \sigma \|Hv_{m,\sigma}\|_2 &\geq c_v, \\ \text{tr}(P_{m,\sigma} H \Sigma_{m,\sigma} H^\top P_{m,\sigma}) &\leq C_\Sigma. \end{aligned} \quad (31)$$

These conditions keep the noise-scaled centered velocity away from zero and uniformly bound the projected posterior covariance. Taking expectations in Eq. (30) and applying Jensen’s inequality gives

$$\begin{aligned} \mathbb{E}_m[\mathcal{I}_m(\sigma)] &\geq \frac{c_v^2}{C_\Sigma} \mathbb{E}_m[\kappa_m(\sigma)^2] \\ &\geq \frac{c_v^2}{C_\Sigma} (\mathbb{E}_m[\kappa_m(\sigma)])^2 \\ &= C_I \bar{\kappa}(\sigma)^2, \end{aligned} \quad (32)$$

where $C_I = c_v^2/C_\Sigma$ and $\bar{\kappa}(\sigma) = \mathbb{E}_m[\kappa_m(\sigma)]$. This proves Eq. (9).

Connection to the empirical statistic. The analysis assumes the exact conditional velocity field and its continuous flow trajectories. In practice, we estimate turning from discrete states along numerical trajectories of the CFG-guided teacher field, which need not coincide with the exact conditional field. The proposition therefore motivates the empirical turning statistic without establishing the exact solution.

D. Diagnostic Protocols

D.1. Representations for temporal affinity

Protocol. We compare VAE and V-JEPA temporal affinities A_i on 32 frozen student rollouts with paired targets. Both spaces use Eq. (12) with lags $\{1, 2, 4\}$ and weights $(0.2, 0.3, 0.5)$.

As an independent reference, we use CoTracker3 [17] to track selected target-video points in each generated video, initialized at corresponding first-frame coordinates. We normalize 2-D point displacements by image width and height and compare them at matched relative times. Using the same temporal lags, weights, and similarity formula as for A_i , we obtain a trajectory-agreement score T_i based on point motion rather than feature changes. Spearman ρ compares the rankings of A_i and T_i . Generator quartile recall measures how many of the eight lowest- T_i rollouts are recovered by the eight lowest- A_i rollouts.

Results. Figure 7(a) plots temporal-affinity and trajectory-agreement ranks for all 32 pairs. VAE/V-JEPA affinities attain Spearman correlations of 0.162/0.374, Kendall correlations of 0.117/0.246, and quartile recalls of 25%/50%.

Table S2. **Difficulty ranking by the trained predictor.** Predictor ρ is the Spearman correlation between predicted fitting difficulty $\hat{r}_i$ and measured critic FM loss on 32 fresh rollouts. FM losses average three independent noise draws at each noise level.

Noise σ	0.5	0.8	0.97
Predictor ρ	0.758	0.898	0.952

D.2. Motion magnitude and critic fitting difficulty

Measurement protocol. We evaluate 32 frozen rollouts with the step-1500 critic at $\sigma \in \{0.5, 0.8, 0.92, 0.97\}$. FM loss is the mean squared velocity error over future latent blocks, averaged over three independent re-noising draws per rollout and noise level. We measure latent motion magnitude by averaging RMS latent differences over time and lags $\{1, 2, 4, 8\}$.

Offline representation probes. We fit two-hidden-layer MLPs to predict log-FM loss from VAE and V-JEPA temporal descriptors, using four-fold cross-validation grouped by rollout. Standardization and PCA are fitted on training folds only. Out-of-fold Spearman correlations are computed within each noise level and averaged over all evaluated levels. VAE descriptors obtain 0.460 ± 0.048 , compared with -0.254 ± 0.125 for V-JEPA temporal features. Their hardest-quartile recalls are $36.7 \pm 5.9\%$ and $12.5 \pm 5.7\%$, respectively (chance: 25%).

D.3. Difficulty ranking by the trained predictor

We evaluate the online-trained predictor from the frozen step-2000 DyMD checkpoint on 32 fresh rollouts generated with training-time random denoising exits. At each noise level, Spearman ρ compares predicted fitting difficulty $\hat{r}_i$ with critic FM loss averaged over three independent re-noising draws per rollout. Table S2 reports correlations of 0.758–0.952, showing that the trained predictor captures rollout-dependent fitting-difficulty rankings.

E. Objective and Optimization Details

E.1. Dynamics-guided fake-score tracking

For detached student latents X_i , we form RMS-difference sequences $q_{i,j}^{(\ell)} = \text{RMS}(X_{i,j+\ell} - X_{i,j})$ at lags $\ell \in \{1, 2, 4, 8\}$. The fifth sequence measures RMS latent differences relative to the first block. Each sequence q is summarized by $S(q) = [\text{mean}(q), \text{std}(q), \text{max}(q), q_{\text{last}}]$. We concatenate the summaries in lag order 1, 2, 4, 8, followed by the first-block displacement summary, into a 20-dimensional vector. Transforming each component x to $\log(1 + x)$ gives the descriptor d_i . The two-hidden-layer MLP receives d_i together with the fixed Fourier noise-level embedding $e(t_i)$ to predict fitting difficulty $\hat{r}_i$.

Table S3. Default hyperparameters for re-noise sampling and fake-score tracking.

Quantity	Choice
Latent measurement / spatial size	21 blocks
V-JEPA input / output blocks	16 frames / 8 blocks
Temporal-affinity lags / weights	$\{1, 2, 4\}$ / $(0.2, 0.3, 0.5)$
Teacher-prior probability floor α	0.3
Turning-rate offset ϵ_κ	0.1 median($\tilde{\kappa}$)
Critic timestep sampling	Same rule as generator
Latent descriptor lags	$\{1, 2, 4, 8\}$
Descriptor dimension	20
Critic residual ema decay ρ_b	0.99
Difficulty predictor	Two-hidden-layer MLP
Critic-weighting strength β	0.5
Predictor loss weight λ_{pred}	1.0

Using the per-rollout mean squared FM loss over future latent blocks, the noise-bin baseline and predictor supervision are updated as

$$\begin{aligned}
b_k &\leftarrow \rho_b b_k + (1 - \rho_b) \text{mean}_{i:k(t_i)=k} \log(\ell_i^{\text{FM}} + \epsilon_c), \\
r_i &= \text{sg}[\log(\ell_i^{\text{FM}} + \epsilon_c) - b_{k(t_i)}], \\
\mathcal{L}_{\text{pred}} &= \frac{1}{B} \sum_{i=1}^B \text{Huber}(\hat{r}_i, r_i).
\end{aligned} \tag{33}$$

Here ρ_b is the EMA decay and $\epsilon_c > 0$ stabilizes the logarithm. Bin statistics are aggregated across workers; bins without samples retain their previous baseline.

Table S3 collects the default hyperparameters used.

F. Benchmark Protocols and Metrics

R-Bench. We evaluate on 400 image-prompt pairs from the embodiment split of R-Bench [8], spanning four robot embodiments (dual-arm, humanoid, single-arm, and quadruped). GPT-5 assigns each video a 1–5 task-completion rating r_v from a grid of six uniformly sampled frames. We normalize scores as $(r_v - 1)/4$ and compute Overall by averaging first within each embodiment, then equally across the four embodiments.

PAI-Bench-G. We evaluate the 174-condition robot subset of PAI-Bench-G [38], comprising 913 binary questions. Eight frames are sampled uniformly and judged with Qwen3-VL-235B using the official binary-VQA templates. Domain averages per-video accuracies, while Space, Physics, and Time micro-average question accuracies within each axis, assessing spatial interactions, physical plausibility, and temporal progression, respectively.

PAI Quality is the mean of eight normalized dimensions: subject consistency, background consistency, motion smoothness, aesthetic quality, imaging quality, overall consistency,

Table S4. **PAI-Bench-G Quality breakdown.** Quality is the unweighted mean of the eight official metrics. SC: subject consistency; BC: background consistency; MS: motion smoothness; AQ: aesthetic quality; IQ: imaging quality; OC: overall consistency; IS/IB: image-to-video subject/background fidelity. All values are percentages; higher is better. Column-wise best values, including ties, are bold and shaded blue.

Method	Quality	SC	BC	MS	AQ
PF-Wan 14B	76.9	90.3	90.6	98.9	50.1
Wan2.1-Fun 1.3B	75.3	87.8	91.6	98.6	47.6
Base DMD	78.1	95.3	91.9	99.5	48.6
DyMD	78.1	95.3	92.2	99.6	48.5
Method	IQ	OC	IS	IB	
PF-Wan 14B	71.6	19.9	96.2	97.5	
Wan2.1-Fun 1.3B	67.2	19.4	94.4	96.0	
Base DMD	73.2	19.5	97.8	98.9	dymd
DyMD	73.1	19.6	97.9	98.9	

Table S5. **Video-native versus framewise temporal-affinity features.** We compare V-JEPA 2.1-L and DINOv2-L for affinity-conditioned sampling under the same training budget. Bold with blue shading marks the better result.

Encoder	R-Bench	PAI-Bench-G		EZS-Bench
	TAC	Domain	Quality	Domain
V-JEPA 2.1-L	43.5	78.5	78.1	75.1
DINOv2-L	41.8	76.7	77.8	75.0

and I2V subject/background fidelity. Table S4 gives the breakdown.

EZS-Bench. EZS-Bench [7] contains 196 unseen robot-task-scene conditions and 2,272 binary questions. We use the same per-video and per-axis aggregation as PAI-Bench-G. Under the official two-model evaluation protocol, Qwen3-VL-32B-Thinking generates the questions and physics checklists, while Qwen2.5-VL-72B-Instruct scores the videos. Following PhysisForcing [36], we use GPT-5 to compress each original description into a 40–60-word action-centric prompt, using the first-frame image as context. The original questions and checklists remain unchanged for scoring.

F.1. Downstream action-planning protocol

Frozen world-model features. Following WorldArena’s action-planner interface [28], we use a VPP inverse-dynamics head [13] to predict actions from frozen student features. At each planning step, we condition the student on the current image and instruction and extract intermediate DiT features at the initial, highest-noise timestep. Feature extraction requires only a single partial forward pass, without

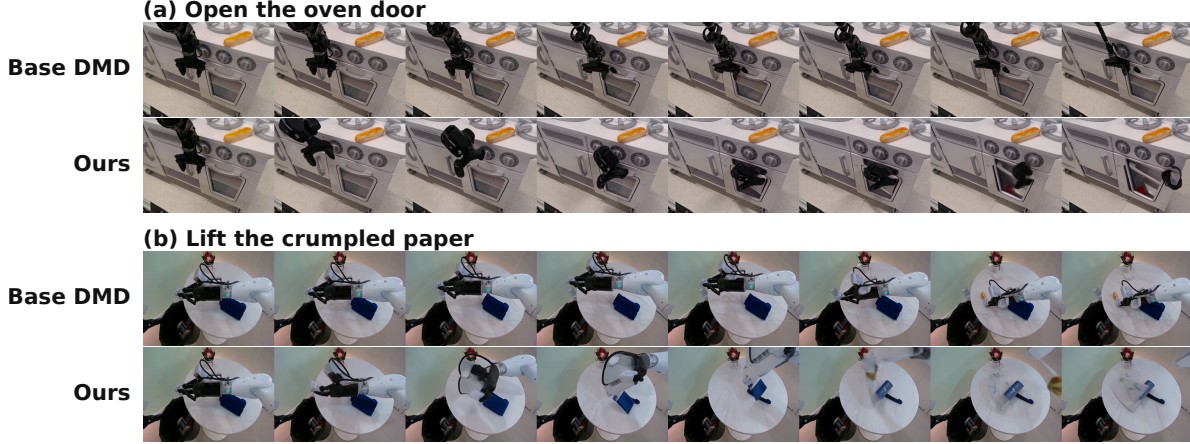


Figure S1. **Additional interaction-dynamics comparisons at four NFE.** Eight uniformly spaced frames are shown for each of two PAI-Bench-G cases, with prompts, seeds, and step-1500 checkpoints matched to the main qualitative results. Base DMD often preserves the initial configuration or incompletely executes the action, whereas DyMD produces clearer object-directed motion.

completing the denoiser or decoding an RGB rollout.

VPP training. We train a separate VPP head for each backbone and task using the 50 demonstrations in `aloha-agilex_clean_50`. Each training sample pairs an observation with a 45-step sequence of 14-dimensional joint actions. The input features span 21 latent temporal blocks, corresponding to an 81-frame horizon. VPP uses a six-layer Video Former with eight attention heads and 399 latent queries, followed by an action-denoising head.

We train for 200 epochs with AdamW, using a learning rate of 10^{-4} , weight decay of 0.05, and $(\beta_1, \beta_2) = (0.9, 0.9)$, with eight processes and a batch size of eight per process. For Adjust Bottle, we evaluate the Base DMD and DyMD heads at steps 13904 and 15010, respectively; for Click Bell, the corresponding steps are 4509 and 5319.

Closed-loop evaluation. VPP predicts 45-action chunks with ten action-denoising steps. Actions are executed in RoboTwin 2.0 [6], with updated observations conditioning subsequent predictions. Each backbone is evaluated over 100 episodes per task.

G. Additional Ablation Studies

G.1. Video-native versus framewise temporal-affinity features

Table S5 compares video-native V-JEPA 2.1-L with frame-wise DINOv2-L [26] for temporal affinity-conditioned re-noise sampling. For DINOv2, we encode 16 uniformly sampled frames into 24×32 patch grids, apply spatial pooling, and average adjacent frame pairs into eight temporal blocks. Relative to DINOv2, V-JEPA improves TAC, PAI Domain, Quality, and EZS Domain by 1.7, 1.8, 0.3, and

0.1 points, respectively, supporting video-native features for target-relative interaction assessment.

G.2. Teacher-prior probability floor

Table S6 varies $\alpha \in \{0.1, 0.3, 0.5\}$ with tracking disabled. Increasing α raises the minimum teacher-prior sampling probability, allocating more samples under the teacher prior schedule. The weaker results at $\alpha = 0.1$ suggest that these intervals receive insufficient supervision: increasing α to 0.3 improves TAC, PAI Domain, and EZS Domain by 4.3, 1.1, and 1.9 points, respectively. Further increasing α to 0.5 yields comparable performance, with TAC increasing by 0.3 points and both Domain scores decreasing by 0.3 points. We therefore use the moderate floor $\alpha = 0.3$, which provides sufficient teacher-prior coverage

Table S6. **Sensitivity to the teacher-prior probability floor α .** Only α varies, with tracking disabled. The $\alpha = 0.3$ row repeats the sampling-only result from Table 2. Column-wise best values, including ties, are bold and shaded blue.

α	R-Bench	PAI-Bench-G		EZS-Bench
	TAC	Domain	Quality	Domain
0.1	39.2	77.4	78.1	73.2
0.3	43.5	78.5	78.1	75.1
0.5	43.8	78.2	78.1	74.8